To: Audit Senior
From: Hamant Rai (Analyst, Quantfolio Solutions)
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RE: Bibitor LLC: Inventory Valuation Risk Assessment

## Introduction

Bibitor LLC may be carrying inventory at values that exceed net realizable value, creating a risk of material misstatement on the balance sheet. Under Bibitor's weighted average cost policy and GAAP, inventory must be recorded at the lower of cost or NRV, and a failure to write down inventory where cost exceeds selling price would overstate assets. Our analysis identified a potential inventory overstatement of $1,905,046.63 across 2,503 brands, which is likely to exceed the engagement's materiality threshold and require a correcting journal entry before audited financial statements can be issued.

## Scope

Sales and ending inventory data were pulled from two client provided datasets covering Bibitor's full brand portfolio across all store locations for the fiscal year. NRV was approximated as the average selling price per brand, consistent with Bibitor's policy. The analysis was scoped to brands where average selling price fell below average purchase cost. A left join between sales and ending inventory was applied on Brand and Store, meaning any brand and store combination absent from the ending inventory snapshot was excluded from the misstatement calculation. Python with pandas was used in Google Colab.

## Methodology

Pricing consistency was first assessed by grouping all sales transactions by brand and calculating average, minimum, and maximum selling prices to identify brands where price varied across transactions during the year. Sales and ending inventory data were then merged on Brand and Store to attach purchase cost to each sales record, enabling a direct cost versus price comparison. Both datasets were aggregated to the brand level, calculating average selling price and average purchase cost per brand. The difference between average cost and average price was multiplied by units on hand at year end for each brand where cost exceeded price, and the results were summed to produce the total potential misstatement.

## Key Findings

Pricing inconsistency was identified across 5,127 of Bibitor's 10,473 brands, approximately 49%, where selling price varied across transactions during the year, indicating pricing controls may not be consistently applied. Of the 10,473 brands, 2,503 had an average selling price below average purchase cost, indicating these brands are potentially carried above NRV. The total potential inventory overstatement is $1,905,046.63, calculated as the cost to price delta multiplied by units on hand at year end. This estimate relies on brand level averages as proxies, and store level or transaction level variation may cause the true figure to differ. Given the scale of the amount, it is unlikely to fall within a standard materiality threshold.

## Recommendations

Management should be informed of the findings before any adjustment is booked. As a verification step, substantive testing on the highest dollar misstatement brands would confirm whether the average cost and price figures accurately reflect the specific units on hand at year end. If verified, a journal entry recognizing an inventory write down for the affected brands should be recorded prior to financial statement issuance. Going forward, a recurring dashboard monitoring cost versus price by brand at each reporting period would allow Bibitor to identify and address NRV concerns on a timely basis rather than at year end.